Computer Science Practical List 3

1. Define the following user-defined functions
   a. GetStory(), to create a text file named STORY.TXT, which should contain text lines entered by the user.
   b. ViewStory(), to read and display the entire content of text file STORY.TXT on the screen.
   c. Words(), to read and display each word of the text file STORY.TXT in different lines on the screen.
   d. AllWords(), to read, count and display number of words present in the text file STORY.TXT

   Also, write menu driven code to call the above functions using a loop.

2. Write a program with the following user-defined functions,
   a. CreateNotes() - to create a text file NOTES.TXT to have text entered by user as lines of text
   b. CountAVD(), to read the contents of the text file NOTES.TXT and count and print the number of Alphabets, Vowels and Digits present in the file.
   c. ShowNotes(), to display the contents of text file NOTES.TXT
   d. RevText(), to read the contents of the text file NOTES.TXT, and print the reverse of those lines which start with an alphabet 'T'.
   e. CountLines(), to read, count and display the number of lines present in the text file NOTES.TXT

   Also, write menu driven code to call the above functions using a loop.

3. Write a program with the following user-defined functions,
   a. Add2Diary() - to create a text file DIARY.TXT to have text entered by the user as lines of text with each line starting with a date in the format DD-Mon-YYYY followed by some activity of the day. A sample content is as follows:

```
23-Jun-2024 Played football, Revised Physics Chapter 1, Visited Uncle Ram
24-Jun-2024 Played football, Revised Maths Chapter 1, Read Lesson 3 English
25-Jun-2024 Played badminton, Revised Chemistry Chapter 2, Read Lesson 2 of English
04-Jul-2024 Played cricket, Revised Chemistry Chapter 3, Read Lesson 4 of English
05-Jul-2024 Played badminton, Revised Chemistry Chapter 4, Read Lesson 5 of English
```

   b. MonthlyActs(Mon), to read the contents of text file DIARY.TXT and display all the activities for every date of the month which is passed to the parameter Mon. For example, if the function is called as follows:

      MonthlyActs("Jun")

      then the output should appear as follows:

```
23-Jun-2024 Played football, Revised Physics Chapter 1, Visited Uncle Ram
24-Jun-2024 Played football, Revised Maths Chapter 1, Read Lesson 3 English
25-Jun-2024 Played badminton, Revised Chemistry Chapter 2, Read Lesson 2 of English
```

   c. ShowRevision(Topic), to display the contents of text file DIARY.TXT and display the number of times the Topic passed to the parameter appears in the file. For example, if the function is called as follows:

      ShowRevision("badminton")

      Then the output should should appear as follows:

      badminton occurs 2 times

   d. LastAct(), to read the contents of the text file DIARY.TXT, and display the last activities of every date (last activities of every date appear after the last comma in each line in the file). For example, if the function is called then the output should appear as follows:

      Visited Uncle Ram
      Read Lesson 3 of English
      Read Lesson 2 of English
      Read Lesson 4 of English
      Read Lesson 5 of English

   Also, write menu driven code to call the above functions using a loop.

4. Write a program to perform the following operations using user defined functions on a text file CAPITALS.TXT
    a. Create() - Create a dictionary named SandC and allow the user to enter N number of Indian states as Keys and their Capitals as the corresponding values. Display the content of SandC and confirm if the user wishes to transfer the content to the text file, if confirmed, allow him/her to store the content of dictionary into the text file CAPITALS.TXT such that the name of every State and its corresponding Capital gets stored in each line of the file separated by a symbol *
    b. Check(), to ask the user to enter a State name and display its corresponding Capital after searching out the State name from the text file CAPITALS.TXT.
    c. Quiz(), to read the text file CAPITALS.TXT and display the State names in random order. After a State name is displayed the user should be asked to enter its corresponding capital name. If entered correctly the user should be awarded a +1 as his score. The task should repeat for the user for 5 correct responses or till the user enters three wrong answers (whichever first). The function should finally display the total score obtained by the user.

    Also, write menu driven code to call the above functions using a loop.
5. Write a program with the following user-defined functions:
    a. CreateStory() - to create a text file STORY.TXT to have multiple lines of text entered by user
    b. BigLines(), to read the contents of the text file STORY.TXT and display only those lines which have more than 80 characters.
    c. TranDelhi(), to create a new file DELHI.TXT which should contain all such lines from the file STORY.TXT which have the word "DELHI" present in it (in upper or lowercase letters)
    d. Recreate(), to read the contents of the text file STORY.TXT, and create another file NEWSTORY.TXT which should not contain the lines carrying the word DELHI present in it (in upper or lowercase letters). Copy the content of NEWSTORY.TXT back into STORY.TXT as fresh content and remove the content of NEWSTORY.TXT

    Also, write menu driven code to call the above functions using a loop.
6. Write a program with the following user-defined functions:
    a. CreateLog() - to create a text file LOG.TXT to keep text one liner write up of each job done by an employee one after another. (The function should not re-create the file, if it already has some content in it, rather it should allow the user to add new lines below the existing lines in the file.)
    b. CompletedWork(), to read the contents of the text file LOG.TXT and display only those lines which contain a word(s) as "Completed" (in any type cases)
    c. CountPending(), to count the number of lines from LOG.TXT, which carry any of the following words - "Doing","Continuing","Halfway" (in any type cases)
    d. ShowLog() - to display entire content of LOG.TXT

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN ASSIGNMENT BOOKLET

General Instructions:

i. Type the solution of all the problems using **Python** Language on **Python IDLE** or **Colab**

ii. Type in the following on top of each of your program as comment lines

```
# ------------------------------------------------------------#
# List-Program No     : L3-P1
# Developed By        : Reema Jones
# Date                : 31-July-2025
# ------------------------------------------------------------#
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. **"Turn in"** the softcopy of the programs as a single document in response to this assignment submission with each program on separate pages. Once verified by the teacher, take out a hardcopy of **each program on separate pages** from the printer and get them signed by the respective computer teacher. Add all the printouts in the practical file and get the Index entry also signed.

v. Recommended Font, "**Courier New**", Style "**Bold**" Size "**10**" for all programs with single line spacing. Default indentation **2 Character** only.

```
# ------------------------------------------------------------#
# List-Program No : L3-P1
# Developed By    : Puja Dey
# Date            : 31-July-2025
# ------------------------------------------------------------#
def GetStory():
    f = open("STORY.TXT", "w")
    while True:
        line = input("Enter line: ")
        f.write(line + "\n")
        ch = input("Want to add more lines? (Y/N): ")
        if ch.upper() == "N":
            break
    f.close()

def ViewStory():
    f = open("STORY.TXT", "r")
    data = f.read()
    print(data)
    f.close()

def Words():
    f = open("STORY.TXT", "r")
    for line in f:
        for word in line.split():
            print(word)
    f.close()

def AllWords():
    f = open("STORY.TXT", "r")
    count = 0
    for line in f:
        words = line.split()
        count += len(words)
    print("Number of words:", count)
    f.close()

while True:
    print("\n1. GetStory\n2. ViewStory\n3. Words\n4. AllWords\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        GetStory()
    elif ch == 2:
```

```
        ViewStory()
    elif ch == 3:
        Words()
    elif ch == 4:
        AllWords()
    elif ch == 5:
        break
```

```
1. GetStory
2. ViewStory
3. Words
4. AllWords
5. Exit
Enter choice: 1
Enter line: Once upon a time in Delhi
Want to add more lines? (Y/N): Y
Enter line: It was a bright sunny day
Want to add more lines? (Y/N): Y
Enter line: And I was feeling sleepy.
Want to add more lines? (Y/N): N

1. GetStory
2. ViewStory
3. Words
4. AllWords
5. Exit
Enter choice: 2
Once upon a time in Delhi
It was a bright sunny day
And I was feeling sleepy.


1. GetStory
2. ViewStory
3. Words
4. AllWords
5. Exit
Enter choice: 3
Once
upon
a
time
in
Delhi
It
was
a
bright
sunny
day
And
I
was
feeling
sleepy.
```

```
1. GetStory
2. ViewStory
3. Words
4. AllWords
5. Exit
Enter choice: 4
Number of words: 17

1. GetStory
2. ViewStory
3. Words
4. AllWords
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L3-P2
# Developed By    : Puja Dey
# Date            : 31-July-2025
# ------------------------------------------------------------#

def CreateNotes():
    f = open("NOTES.TXT", "w")
    while True:
        line = input("Enter line: ")
        f.write(line + "\n")
        ch = input("Want to add more lines? (Y/N): ")
        if ch.upper() == "N":
            break
    f.close()

def CountAVD():
    f = open("NOTES.TXT", "r")
    text = f.read()
    alphabets = vowels = digits = 0
    for ch in text:
        if ch.isalpha():
            alphabets += 1
            if ch.lower() in "aeiou":
                vowels += 1
        elif ch.isdigit():
            digits += 1
    f.close()
    print("Alphabets:", alphabets)
    print("Vowels:", vowels)
    print("Digits:", digits)

def ShowNotes():
    f = open("NOTES.TXT", "r")
    print(f.read())
    f.close()

def RevText():
    f = open("NOTES.TXT", "r")
    for line in f:
        if line[0].upper() == "T":
            print(line[::-1].strip())
    f.close()
```

```
def CountLines():
    f = open("NOTES.TXT", "r")
    count = 0
    for line in f:
        count += 1
    f.close()
    print("Number of lines:", count)

while True:
    print("\n1. CreateNotes\n2. CountAVD\n3. ShowNotes\n4. RevText\n5. CountLines\n6.
     Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        CreateNotes()
    elif ch == 2:
        CountAVD()
    elif ch == 3:
        ShowNotes()
    elif ch == 4:
        RevText()
    elif ch == 5:
        CountLines()
    elif ch == 6:
        break
```

```
1. CreateNotes
2. CountAVD
3. ShowNotes
4. RevText
5. CountLines
6. Exit
Enter choice: 1
Enter line: Today is 2nd April 2025
Want to add more lines? (Y/N): Y
Enter line: The event starts at 10am
Want to add more lines? (Y/N): Y
Enter line: Student scored 100 marks in CS
Want to add more lines? (Y/N): N

1. CreateNotes
2. CountAVD
3. ShowNotes
4. RevText
5. CountLines
6. Exit
Enter choice: 2
Alphabets: 54
Vowels: 17
Digits: 10
```

```
1. CreateNotes
2. CountAVD
3. ShowNotes
4. RevText
5. CountLines
6. Exit
Enter choice: 3
Today is 2nd April 2025
The event starts at 10am
Student scored 100 marks in CS


1. CreateNotes
2. CountAVD
3. ShowNotes
4. RevText
5. CountLines
6. Exit
Enter choice: 4
5202 lirpA dn2 si yadoT
ma01 ta strats tneve ehT

1. CreateNotes
2. CountAVD
3. ShowNotes
4. RevText
5. CountLines
6. Exit
Enter choice: 5
Number of lines: 3

1. CreateNotes
2. CountAVD
3. ShowNotes
4. RevText
5. CountLines
6. Exit
Enter choice: 6
```

```
# ------------------------------------------------------------#
# List-Program No : L3-P3
# Developed By    : Puja Dey
# Date            : 31-July-2025
# ------------------------------------------------------------#
def Add2Diary():
    f = open("DIARY.TXT", "w")
    while True:
        line = input("Enter line (DD-Mon-YYYY Activity...): ")
        f.write(line + "\n")
        ch = input("Want to add more lines? (Y/N): ")
        if ch.upper() == "N":
            break
    f.close()

def MonthlyActs(Mon):
    f = open("DIARY.TXT", "r")
    for line in f:
        if "-" + Mon + "-" in line:
            print(line.strip())
    f.close()

def ShowRevision(Topic):
    f = open("DIARY.TXT", "r")
    count = 0
    for line in f:
        if Topic.lower() in line.lower():
            count += line.lower().count(Topic.lower())
    f.close()
    print(Topic, "occurs", count, "times")

def LastAct():
    f = open("DIARY.TXT", "r")
    for line in f:
        parts = line.strip().split(",")
        print(parts[-1].strip())
    f.close()

while True:
    print("\n1. Add2Diary\n2. MonthlyActs\n3. ShowRevision\n4. LastAct\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        Add2Diary()
```

```
    elif ch == 2:
        Mon = input("Enter month (e.g. Jun, Jul): ")
        MonthlyActs(Mon)
    elif ch == 3:
        Topic = input("Enter topic to search: ")
        ShowRevision(Topic)
    elif ch == 4:
        LastAct()
    elif ch == 5:
        break
```

```
1. Add2Diary
2. MonthlyActs
3. ShowRevision
4. LastAct
5. Exit
Enter choice: 1
Enter line (DD-Mon-YYYY Activity...): 23-Jun-2024 Played football, Revised Physics
      Chapter 1, Visited Uncle
Want to add more lines? (Y/N): Y
Enter line (DD-Mon-YYYY Activity...): 24-Jun-2024 Played football, Revised Maths
      Chapter 1, Read Lesson 3 English
Want to add more lines? (Y/N): Y
Enter line (DD-Mon-YYYY Activity...): 25-Jun-2024 Played badminton, Revised Chemistry
      Chapter 2, Read Lesson 2 of English
Want to add more lines? (Y/N): Y
Enter line (DD-Mon-YYYY Activity...): 04-Jul-2024 Played cricket, Revised Chemistry
      Chapter 3, Read Lesson 4 of English
Want to add more lines? (Y/N): Y
Enter line (DD-Mon-YYYY Activity...): 05-Jul-2024 Played badminton, Revised Chemistry
      Chapter 4, Read Lesson 5 of English
Want to add more lines? (Y/N): N

1. Add2Diary
2. MonthlyActs
3. ShowRevision
4. LastAct
5. Exit
Enter choice: 2
Enter month (e.g. Jun, Jul): Jun
23-Jun-2024 Played football, Revised Physics Chapter 1, Visited Uncle
24-Jun-2024 Played football, Revised Maths Chapter 1, Read Lesson 3 English
25-Jun-2024 Played badminton, Revised Chemistry Chapter 2, Read Lesson 2 of English

1. Add2Diary
2. MonthlyActs
3. ShowRevision
4. LastAct
5. Exit
Enter choice: 3
Enter topic to search: English
English occurs 4 times

1. Add2Diary
2. MonthlyActs
```

```
3. ShowRevision
4. LastAct
5. Exit
Enter choice: 4
Visited Uncle
Read Lesson 3 English
Read Lesson 2 of English
Read Lesson 4 of English
Read Lesson 5 of English

1. Add2Diary
2. MonthlyActs
3. ShowRevision
4. LastAct
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L3-P4
# Developed By    : Puja Dey
# Date            : 31-July-2025
# ------------------------------------------------------------#

import random

def Create():
    SandC = {}
    n = int(input("Enter number of states: "))
    for i in range(n):
        state = input("Enter state: ")
        capital = input("Enter capital: ")
        SandC[state] = capital
    print("Dictionary:", SandC)
    ch = input("Do you want to save into CAPITALS.TXT? (Y/N): ")
    if ch.upper() == "Y":
        f = open("CAPITALS.TXT", "w")
        for s in SandC:
            f.write(s + "*" + SandC[s] + "\n")
        f.close()

def Check():
    state = input("Enter state name to search: ")
    f = open("CAPITALS.TXT", "r")
    found = False
    for line in f:
        s, c = line.strip().split("*")
        if s.lower() == state.lower():
            print("Capital of", s, "is", c)
            found = True
            break
    f.close()
    if not found:
        print("State not found")

def Quiz():
    f = open("CAPITALS.TXT", "r")
    pairs = []
    for line in f:
        s, c = line.strip().split("*")
        pairs.append((s, c))
```

```
    f.close()
    random.shuffle(pairs)
    score = 0
    wrong = 0
    i = 0
    while score < 5 and wrong < 3 and i < len(pairs):
        state, capital = pairs[i]
        ans = input("Enter capital of " + state + ": ")
        if ans.lower() == capital.lower():
            print("Correct!")
            score += 1
        else:
            print("Wrong! Correct is", capital)
            wrong += 1
        i += 1
    print("Final Score:", score)

while True:
    print("\n1. Create\n2. Check\n3. Quiz\n4. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        Create()
    elif ch == 2:
        Check()
    elif ch == 3:
        Quiz()
    elif ch == 4:
        break
```

```
1. Create
2. Check
3. Quiz
4. Exit
Enter choice: 1
Enter number of states: 5
Enter state: West Bengal
Enter capital: Kolkata
Enter state: Maharashtra
Enter capital: Mumbai
Enter state: Tamil Nadu
Enter capital: Chennai
Enter state: Uttar Pradesh
Enter capital: Lucknow
Enter state: Karnataka
Enter capital: Bengaluru
Dictionary: {'West Bengal': 'Kolkata', 'Maharashtra': 'Mumbai', 'Tamil Nadu':
      'Chennai', 'Uttar Pradesh': 'Lucknow', 'Karnataka': 'Bengaluru'}
Do you want to save into CAPITALS.TXT? (Y/N): Y
```

```
1. Create
2. Check
3. Quiz
4. Exit
Enter choice: 2
Enter state name to search: Delhi
State not found

1. Create
2. Check
3. Quiz
4. Exit
Enter choice: 2
Enter state name to search: Maharashtra
Capital of Maharashtra is Mumbai

1. Create
2. Check
3. Quiz
4. Exit
Enter choice: 3
Enter capital of Karnataka: New Delhi
Wrong! Correct is Bengaluru
Enter capital of West Bengal: Kolkata
Correct!
Enter capital of Uttar Pradesh: Lucknow
Correct!
Enter capital of Tamil Nadu: Hyderabad
Wrong! Correct is Chennai
Enter capital of Maharashtra: Bombay
Wrong! Correct is Mumbai
Final Score: 2

1. Create
2. Check
3. Quiz
4. Exit
Enter choice: 4
```

```
# ------------------------------------------------------------#
# File-Program No  : L3-P5
# Developed By     : PUJA DEY
# Date             : 14-April-2025
# ------------------------------------------------------------#

def CreateStory():
    f = open("STORY.TXT", "w")
    while True:
        line = input("Enter line: ")
        f.write(line + "\n")
        ch = input("Want to add more lines? (Y/N): ")
        if ch.upper() == "N":
            break
    f.close()

def BigLines():
    f = open("STORY.TXT", "r")
    for line in f:
        if len(line.strip()) > 80:
            print(line.strip())
    f.close()

def TranDelhi():
    f1 = open("STORY.TXT", "r")
    f2 = open("DELHI.TXT", "w")
    for line in f1:
        if "delhi" in line.lower():
            f2.write(line)
    f1.close()
    f2.close()

def Recreate():
    f1 = open("STORY.TXT", "r")
    f2 = open("NEWSTORY.TXT", "w")
    for line in f1:
        if "delhi" not in line.lower():
            f2.write(line)
    f1.close()
    f2.close()
    f2 = open("NEWSTORY.TXT", "r")
    data = f2.read()
```

```
    f2.close()
    f3 = open("STORY.TXT", "w")
    f3.write(data)
    f3.close()
    f4 = open("NEWSTORY.TXT", "w")
    f4.close()

while True:
    print("\n1. CreateStory\n2. BigLines\n3. TranDelhi\n4. Recreate\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        CreateStory()
    elif ch == 2:
        BigLines()
    elif ch == 3:
        TranDelhi()
        f = open("DELHI.TXT", "r")
        print("Content of DELHI.TXT:")
        print(f.read())
        f.close()
    elif ch == 4:
        Recreate()
        f = open("STORY.TXT", "r")
        print("Updated STORY.TXT:")
        print(f.read())
        f.close()
    elif ch == 5:
        break
```

```
1. CreateStory
2. BigLines
3. TranDelhi
4. Recreate
5. Exit
Enter choice: 1
Enter line: Delhi is the capital of India.
Want to add more lines? (Y/N): Y
Enter line: The Yamuna river flows through Delhi and the city has a rich history with
      monuments like Red Fort and Qutub Minar which attract many tourists every year.
Want to add more lines? (Y/N): Y
Enter line: Python programming is very powerful and easy to learn.
Want to add more lines? (Y/N): N

1. CreateStory
2. BigLines
3. TranDelhi
4. Recreate
5. Exit
Enter choice: 2
```

```
The Yamuna river flows through Delhi and the city has a rich history with monuments
      like Red Fort and Qutub Minar which attract many tourists every year.

1. CreateStory
2. BigLines
3. TranDelhi
4. Recreate
5. Exit
Enter choice: 3
Content of DELHI.TXT:
Delhi is the capital of India.
The Yamuna river flows through Delhi and the city has a rich history with monuments
      like Red Fort and Qutub Minar which attract many tourists every year.


1. CreateStory
2. BigLines
3. TranDelhi
4. Recreate
5. Exit
Enter choice: 4
Updated STORY.TXT:
Python programming is very powerful and easy to learn.


1. CreateStory
2. BigLines
3. TranDelhi
4. Recreate
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L3-P6
# Developed By    : Puja Dey
# Date            : 31-July-2025
# ------------------------------------------------------------#
def CreateLog():
    f = open("LOG.TXT", "a")
    while True:
        line = input("Enter job line: ")
        f.write(line + "\n")
        ch = input("Want to add more lines? (Y/N): ")
        if ch.upper() == "N":
            break
    f.close()
def CompletedWork():
    f = open("LOG.TXT", "r")
    for line in f:
        if "completed" in line.lower():
            print(line.strip())
    f.close()
def CountPending():
    f = open("LOG.TXT", "r")
    count = 0
    for line in f:
        l = line.lower()
        if "doing" in l or "continuing" in l or "halfway" in l:
            count += 1
    f.close()
    print("Pending lines:", count)
def ShowLog():
    f = open("LOG.TXT", "r")
    print(f.read())
    f.close()
while True:
    print("\n1. CreateLog\n2. CompletedWork\n3. CountPending\n4. ShowLog\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        CreateLog()
    elif ch == 2:
        CompletedWork()
    elif ch == 3:
        CountPending()
    elif ch == 4:
```

```
        ShowLog()
    elif ch == 5:
        break
```

```
1. CreateLog
2. CompletedWork
3. CountPending
4. ShowLog
5. Exit
Enter choice: 1
Enter job line: Completed Project Report
Want to add more lines? (Y/N): Y
Enter job line: Doing Data Analysis
Want to add more lines? (Y/N): Y
Enter job line: Continuing Python Assignment
Want to add more lines? (Y/N): Y
Enter job line: Halfway through presentation design
Want to add more lines? (Y/N): Y
Enter job line: Completed Team Presentation
Want to add more lines? (Y/N): N

1. CreateLog
2. CompletedWork
3. CountPending
4. ShowLog
5. Exit
Enter choice: 2
Completed Project Report
Completed Team Presentation

1. CreateLog
2. CompletedWork
3. CountPending
4. ShowLog
5. Exit
Enter choice: 3
Pending lines: 3

1. CreateLog
2. CompletedWork
3. CountPending
4. ShowLog
5. Exit
Enter choice: 4
Completed Project Report
Doing Data Analysis
Continuing Python Assignment
Halfway through presentation design
Completed Team Presentation


1. CreateLog
2. CompletedWork
3. CountPending
4. ShowLog
5. Exit
Enter choice: 5
```